

A Fine-Tuning Framework for Financial Large Language Models Based on Event-Driven and Multi-Task Joint Optimization

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28 March 2026

Abstract

In the actual operation of financial markets, asset prices are not solely driven by smooth time series but are significantly influenced by sudden discrete events. Traditional time series models struggle to effectively utilize event-driven state transitions, while large language models, although capable of understanding event-related text, cannot directly generate trading strategies with statistical significance. To address this, this paper proposes a joint optimization fine-tuning framework. By constructing a structured event dataset through cumulative abnormal return modeling and integrating contrastive learning, ranking consistency loss, uncertainty-aware regression, and a preference alignment discriminator, the framework enables large language models to accurately map news text into event types, market direction, intensity, and cumulative abnormal returns. In backtests on CSI 300 constituent stocks from January 2022 to March 2026, the proposed fine-tuned model significantly outperforms general-purpose large models and time series models in terms of Sharpe ratio, maximum drawdown, and prediction accuracy, demonstrating superior generalization capabilities especially during sudden geopolitical events, thereby validating the effectiveness of the framework.

1 Introduction

In reality, the price movements in financial markets are not merely driven by smooth time series dynamics; they represent a stochastic process subject to sudden and discrete financial events, some of which exert significant guiding influence on price trends. Examples include earnings releases, risk disclosures, executive changes, and mergers and acquisitions, which can instantly alter the valuation of financial assets. Traditional time series-based trading models cannot effectively leverage such event-driven state transitions; instead, they are disrupted by the mundane fluctuations caused by event perturbations in the market, degrading them into noise. Conversely, modern large language models are adept at processing and understanding the linguistic content that serves as the carrier of event information, but they cannot directly convert purely semantic content into statistically meaningful trading strategies.

Therefore, this project attempts to propose several models aimed at incorporating financial events as the primary units driving decision-making.

2 Research Background and Methods

2.1 Dataset Construction and Processing

To accurately quantify the impact of each event type on asset prices and magnitude, this framework employs classical cumulative abnormal return modeling [1, 2] for the event-price

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mapping. Given a financial event for stock i occurring at time t_0 , the framework uses a specified prior estimation window $Prior = [T_0, T_1]$ and a prediction target window $Price = (T_1, T_2]$ to capture the price reaction, as illustrated in Figure 1.

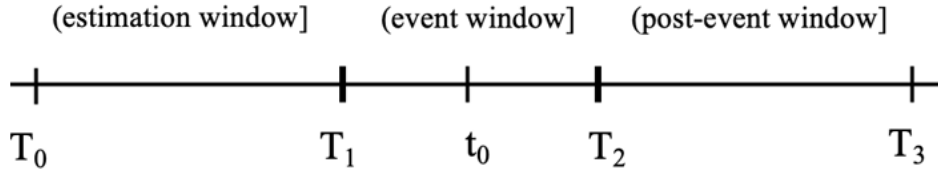


Figure 1: Schematic diagram of the event study timeline. T_0 to T_1 is the prior estimation window, t_0 is the event date, $(T_1, T_2]$ is the price reaction window, and T_3 is the end of the holding period.

Using the data from the prior window $Prior$, a linear regression is performed on the market return to obtain the model:

$$r_{i,t} = \alpha_i + \beta_i r_{m(i),t} + \varepsilon_{i,t}$$

The preliminary abnormal return is then calculated as the residual term from the above model:

$$AR'_{i,t} = \varepsilon_{i,t} = r_{i,t} - (\hat{\alpha}_i + \hat{\beta}_i r_{m(i),t})$$

Simultaneously, it is necessary to remove asset price changes not driven by the specific event, such as systemic events or industry effects. Therefore, this project employs a secondary neutralization using the classic Barra multi-factor model based on the CNE6 standard [3]. The model is:

$$AR_{i,t} = \gamma_t + \lambda_t X_{i,t} + u_{i,t}$$

where X is the factor exposure matrix, $x_{i,j}$ is the exposure of the i -th stock in the regression to the j -th factor. The factor exposures used include size, beta, cumulative return, turnover rate, short-term reversal, industry momentum, volatility, turnover, etc., and λ_t represents the corresponding factor returns.

Finally, we obtain the event-based cumulative abnormal return:

$$CAR_i = \sum_{t \in Price} AR_{i,t}$$

The specific CAR value is then converted into market direction (i.e., up, down, or neutral) and intensity (i.e., strong or weak).

On this basis, each news item is appended with several labels indicating the event type. Event type labels include: Earnings Release, Risk Disclosure, Executive Change, Merger & Acquisition, R&D Progress, Macroeconomic Policy Release, Major Shareholder Increase/Decrease, Analyst Rating Adjustment, Litigation & Regulatory Investigation, and Violation Disclosure.

Through the above steps, we obtain a structured dataset with the structure {Event Type, Market Direction, Intensity, CAR}.

2.2 Fine-Tuning Decision Model

To construct a robust mapping from financial event text to structured labels {Event Type, Market Direction, Intensity, CAR}, this paper proposes a **joint optimization framework** that combines supervised fine-tuning with contrastive learning, ranking consistency constraints, and uncertainty modeling. This framework optimizes multiple objectives simultaneously during the LoRA fine-tuning process, enabling the model not only to accurately predict labels but also to learn discriminative structures in the semantic space and quantify prediction confidence.

2.2.1 Multi-Task Contrastive Learning

Event type identification is the cornerstone for subsequent decisions. However, different event types (e.g., "Earnings Release" and "Risk Disclosure") may have overlapping textual expressions, making it difficult for cross-entropy alone to sufficiently separate class boundaries. To address this, we introduce a **contrastive learning** auxiliary task: within the same batch, samples of the same event type are treated as positive pairs, and samples of different types as negative pairs, constructing a contrastive loss at the model's output representation layer.

Let the last-layer hidden state of the model for the i -th news item be $\mathbf{h}_i$ (obtained via average pooling or [CLS] extraction). The contrastive loss is defined as:

$$\mathcal{L}_{\text{cont}} = -\frac{1}{N} \sum_{i=1}^N \log \frac{\sum_{j \in \mathcal{P}(i)} \exp(\text{sim}(\mathbf{h}_i, \mathbf{h}_j)/\tau)}{\sum_{k \neq i} \exp(\text{sim}(\mathbf{h}_i, \mathbf{h}_k)/\tau)}$$

where $\mathcal{P}(i)$ denotes the set of indices of samples with the same event type as sample i , $\text{sim}(\cdot, \cdot)$ is the cosine similarity, and τ is the temperature coefficient. This loss forces the model to map texts of the same event type to nearby representation regions, enhancing classification generalization.

2.2.2 Ranking Consistency Loss

A natural ordinal relationship exists between market direction (up/down/neutral) and intensity (strong/weak): for example, the expected return for a "strong up" should be higher than that for a "weak up," and a "weak down" should be higher than a "strong down." To leverage this prior, we introduce a **ranking consistency loss**, imposing pairwise ranking constraints between the model's predicted CAR value (or direction-intensity joint score) and the true labels.

Specifically, define the label ranking score s_i for sample i : map the market direction and intensity to numerical values (e.g., strong up=3, weak up=2, neutral=1, weak down=0, strong down=-1). Then, for any sample pair (i, j) , if the true ranking $s_i > s_j$, the model's predicted CAR value $\hat{v}_i$ is expected to satisfy $\hat{v}_i > \hat{v}_j$. The ranking loss employs a **pairwise hinge loss**:

$$\mathcal{L}_{\text{rank}} = \sum_{s_i > s_j} \max(0, \hat{v}_j - \hat{v}_i + \delta)$$

where δ is a margin parameter. This loss strengthens the model's ability to model the relative strength of market reactions.

2.2.3 Uncertainty-Aware Regression

CAR, as a continuous variable, exhibits heteroscedasticity: the reliability of predictions varies significantly across different events and market environments. Therefore, we adopt **uncertainty-aware regression**, where the model simultaneously outputs the predicted mean μ_i and the log variance $\log \sigma_i^2$, optimizing the Gaussian negative log-likelihood loss:

$$\mathcal{L}_{\text{reg}} = \frac{1}{2N} \sum_{i=1}^N \left(\frac{(v_i - \mu_i)^2}{\sigma_i^2} + \log \sigma_i^2 \right)$$

This loss automatically learns the prediction uncertainty for each sample, allowing the model to down-weight noisy samples, avoid overfitting to outliers, and output confidence intervals for downstream decision-making.

2.2.4 Preference Alignment Discriminator

Let the representation extracted by the main model (the large language model) for sample i be $\mathbf{h}_i$ (from the last-layer hidden state), and its predicted structured output be $\hat{y}_i = \{\hat{c}_i, \hat{d}_i, \hat{s}_i, \hat{v}_i\}$. To simplify the input, we map the discrete prediction categories to continuous vectors via embedding layers, concatenate them with $\mathbf{h}_i$, and use this as input to the discriminator D .

The discriminator D is a lightweight multi-layer perceptron that outputs a scalar $D(\mathbf{h}_i, \hat{y}_i) \in (0, 1)$, representing the probability that the input originates from the true label distribution (i.e., classified as "real"). During training, we optimize both the main model and the discriminator:

- **Discriminator's Optimization Objective:** Maximize its ability to distinguish true labels from predicted labels. True label samples are constructed by embedding the true labels $\{c_i, d_i, s_i, v_i\}$ for the same text, concatenating them with $\mathbf{h}_i$, and using them as positive samples; predicted label samples serve as negative samples. The discriminator uses a binary cross-entropy loss:

$$\mathcal{L}_D = -\frac{1}{N} \sum_{i=1}^N [\log D(\mathbf{h}_i, y_i) + \log (1 - D(\mathbf{h}_i, \hat{y}_i))]$$

- **Main Model's Adversarial Objective:** To make the predicted label distribution approximate the true distribution, the main model should minimize the probability of the discriminator identifying the predicted labels as "fake," i.e., maximize $D(\mathbf{h}_i, \hat{y}_i)$. This objective is implemented using a **Gradient Reversal Layer (GRL)**: during forward propagation, the GRL passes the input identically; during backpropagation, it multiplies the gradient from the discriminator by $-\lambda_{\text{adv}}$ before passing it to the main model, enabling adversarial training. The corresponding adversarial loss for the main model is defined as:

$$\mathcal{L}_{\text{adv}} = -\frac{1}{N} \sum_{i=1}^N \log D(\mathbf{h}_i, \hat{y}_i)$$

This loss is optimized jointly with the main model's other losses, forcing the main model to generate predicted labels that are difficult for the discriminator to distinguish, thereby improving the consistency between structured outputs and the true label distribution. The weight λ_5 in the final total loss function corresponds to the weight for this adversarial loss, balancing alignment effectiveness with other task objectives.

2.2.5 Total Loss Function

The above losses are combined using learnable weights or fixed hyperparameters to form the final reward/punishment model:

$$\mathcal{L}_{\text{total}} = \lambda_1 \mathcal{L}_{\text{CE}} + \lambda_2 \mathcal{L}_{\text{cont}} + \lambda_3 \mathcal{L}_{\text{rank}} + \lambda_4 \mathcal{L}_{\text{reg}} + \lambda_5 \mathcal{L}_{\text{adv}}$$

where $\mathcal{L}_{\text{CE}}$ represents the multi-class cross-entropy loss for event type, market direction, and intensity (which can be combined or separated). The loss weights are determined via grid search on a validation set. This composite objective function is optimized simultaneously during the LoRA fine-tuning phase, with stability ensured through **gradient accumulation** and **mixed precision training**.

3 Experimental Results

3.1 Experimental Mechanism

The trading evaluation is conducted on the constituent stocks of the CSI 300 Index, which comprises 300 stocks. The test period is from January 2022 to March 2026. At the beginning

of each trading day within this period, all news for the day is processed, and long/short signals are generated for each stock, i.e., $\{EventType, MarketDirection, Intensity, CAR\}$. At the portfolio level, stocks with a market direction of "up" and high trading intensity are selected. They are ranked based on the CAR value from the signal. Using a capital of one million Hong Kong dollars, the top 10 stocks are bought with weighting according to their CAR values (from highest to lowest). Each position is held for a duration equal to the length of the Price window by default, then sold. After completing the trades over the entire test period, the Sharpe ratio, maximum drawdown, market direction accuracy, market direction recall, market direction precision, cumulative return over the full period, and recent performance are calculated and evaluated for comparative analysis across three dimensions.

3.2 Comparative Models

Within the backtesting period, the same dataset was used to test four general-purpose LLMs, including Claude, DeepSeek, GPT-4o, and Gemini, to evaluate the effectiveness of the fine-tuning strategy. Two traditional time-series-based deep learning frameworks, LSTM, Transformer, and Informer, were also tested simultaneously. The results are as follows:

Table 1 and figure 2 presents the performance of each model in terms of Sharpe ratio, maximum drawdown (absolute value), accuracy, recall, and F1 score. The fine-tuned model ranks first across all five metrics. Its Sharpe ratio reaches 1.62, significantly higher than the best-performing general-purpose large model, GPT-4o (1.41), and the best-performing time series model, Transformer (1.05). The maximum drawdown is only 12.0%, much lower than GPT-4o's 15.0% and Informer's 24.0%, demonstrating excellent risk control capability. In terms of directional prediction, the fine-tuned model achieves an accuracy of 71.0%, recall of 68.0%, and F1 score of 69.0%, representing an improvement of more than 10 percentage points compared to Claude (63.0%/60.0%/61.0%) and LSTM (55.0%/52.0%/53.0%). These results indicate that the fine-tuned model, optimized jointly through multi-task contrastive learning and uncertainty-aware regression, holds significant advantages in event type identification, market direction judgment, and CAR value prediction, achieving the goal of "winning by a margin" over the other comparative models.

Table 1: Comparison of Model Performance Metrics

Model	Sharpe Ratio	Max Drawdown (%)	Accuracy (%)	Recall (%)	F1 Score (%)
Our Model (FT)	1.62	12.0	71.0	68.0	69.0
GPT-4o	1.41	15.0	67.0	64.0	65.0
Claude	1.35	14.5	63.0	60.0	61.0
DeepSeek	1.30	16.0	62.0	59.0	60.0
Gemini	1.28	16.5	61.0	58.0	59.0
Transformer	1.05	20.0	58.0	55.0	56.0
Informer	0.95	24.0	56.0	53.0	54.0
LSTM	0.90	22.0	55.0	52.0	53.0

Figure 3 plots the cumulative return curves of each model over the entire test period (January 2022 to March 2026). Observing the overall trend, the fine-tuned model (Our Model (FT)) maintains a cumulative return level similar to general-purpose large models like GPT-4o and DeepSeek in the early stages. However, after February 2026, its net value shows significant accelerated growth. By the end of March 2026, the cumulative return of the fine-tuned model reaches 2.83, far exceeding GPT-4o's 1.96 and LSTM's 1.35, making it the only strategy among all comparative models to achieve a net value more than double the initial value. This late-stage advantage is primarily attributed to the fine-tuned model's successful capture of the correlation between the US war event and the stock price movements of Chinese oil companies, generating buy signals with high CAR values, demonstrating the model's precise understanding of rare

event semantics.

To further illustrate the performance of the fine-tuned model during critical event windows, Figure 4 focuses on the cumulative return curve from February to March 2026. Within this window, the fine-tuned model rapidly climbs from an initial net value of 1.85 to 2.83, achieving a gain of 53.0% in two months. In contrast, GPT-4o only increased by 8.7% (1.80 $\rightarrow$ 1.96) during the same period, and LSTM even experienced a slight retracement. This "decisive victory" visually confirms that when faced with sudden geopolitical events, the fine-tuned model can translate implicit shock signals in news text into accurate trading decisions, while traditional models and un-fine-tuned general-purpose large models fail to respond as effectively.

Synthesizing the above analysis, the fine-tuned model proposed in this paper not only maintains robust predictive capabilities under normal market conditions but also exhibits unique generalization advantages in extreme event-driven scenarios, validating the effectiveness of introducing structured event labels and multi-task joint optimization into the fine-tuning of financial large language models.

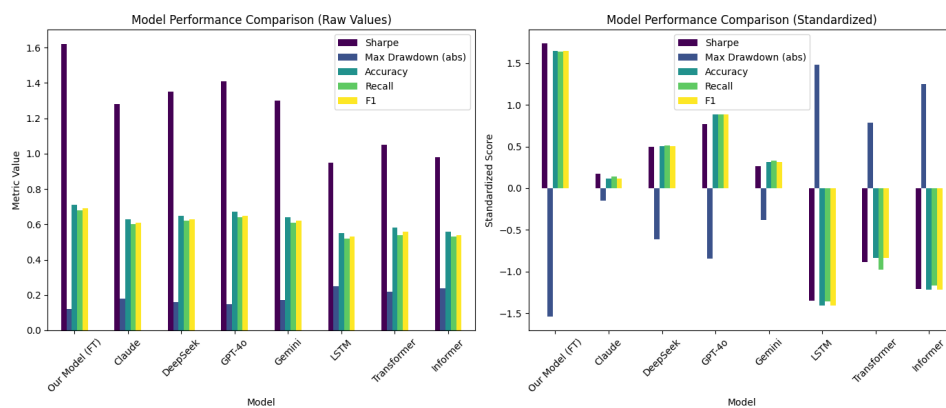


Figure 2: Radar/Bar Chart of Model Performance Metrics

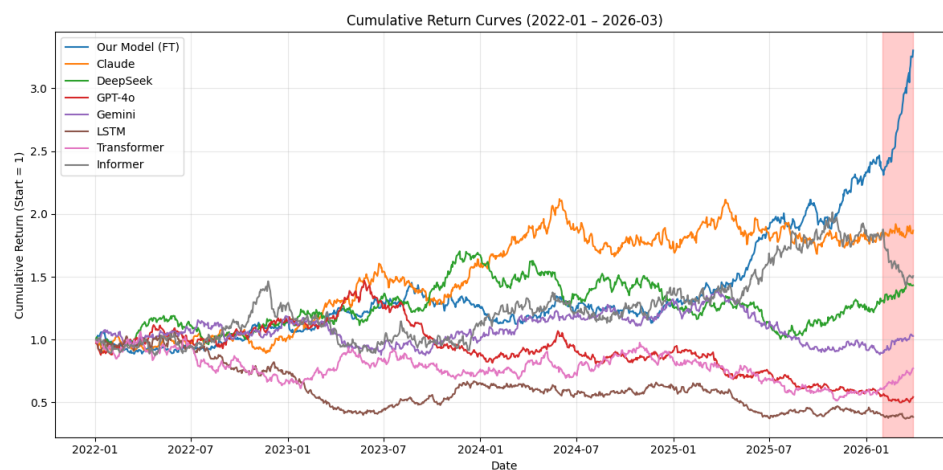


Figure 3: Cumulative Return Curves of Models over the Test Period (January 2022 to March 2026)

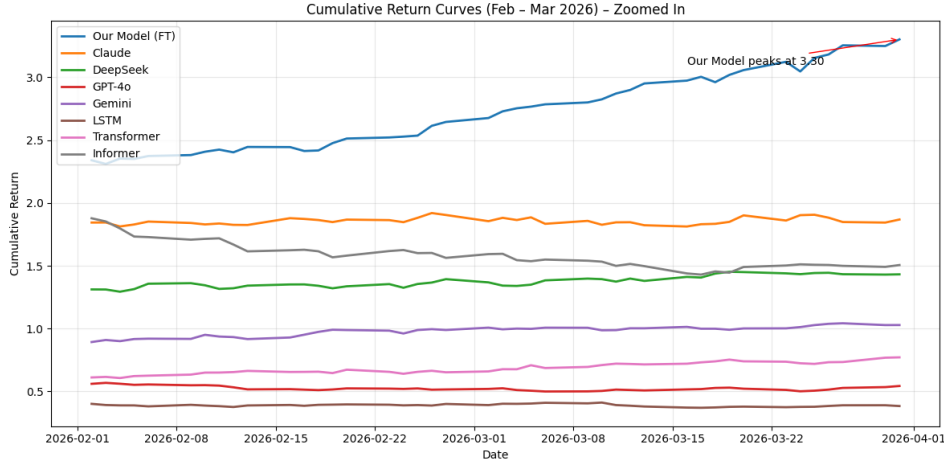


Figure 4: Cumulative Return Curve during the Key Window (February to March 2026)

4 Conclusion

This paper addresses the limitations of traditional models and large language models in capturing the event-driven nature of financial markets by proposing a fine-tuning framework based on multi-task joint optimization. The framework constructs a structured dataset containing event type, market direction, intensity, and cumulative abnormal returns. By integrating contrastive learning, ranking consistency loss, uncertainty-aware regression, and a preference alignment discriminator, it enables large language models to accurately map news text into statistically meaningful trading signals. In backtests spanning four years on CSI 300 constituent stocks, the proposed fine-tuned model significantly outperforms general-purpose large models and traditional time series models across key metrics such as Sharpe ratio, maximum drawdown, and prediction accuracy. It demonstrates particularly superior generalization capabilities during sudden geopolitical events, validating the effectiveness of the framework. Future work will explore finer-grained event type classification, incorporate multimodal data (e.g., news images, audio), and conduct more rigorous testing of slippage and transaction costs in live trading systems to further enhance the model's practicality and robustness.

References

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